

Sign and Conjugate Errors with Complex Numbers

How to work:

- Write every answer in the form $a + bi$ with a and b real, unless the problem asks for something else.
- Remember two facts all the way through: $i^2 = -1$, and the conjugate of $c + di$ is $c - di$ — only the sign in front of the i changes.
- On a “find the mistake” problem, name the wrong move first, then redo the work correctly. Both parts are graded.

1. Subtract and write the result in the form $a + bi$:

$$(3 - 8i) - (5 - 2i)$$

Answer: _____

2. Multiply and write the result in the form $a + bi$:

$$(4 + 3i)(2 - 5i)$$

Answer: _____

3. Multiply the conjugate pair and show why the result has no i in it:

$$(5 + 2i)(5 - 2i)$$

Answer: _____

4. **Find the mistake.** Nadia wants a product that is a real number. She writes

$$(7 - 3i)(7 - 3i) = 58$$

and says, “multiplying by the conjugate clears the i .”

- (a) Explain what went wrong: which factor does the conjugate rule actually call for?
- (b) Compute $(7 - 3i)(7 - 3i)$ correctly.

Answer: _____

- 5.** Divide. Multiply by the conjugate of the denominator and write the result in the form $a + bi$:

$$\frac{8 + i}{2 - 3i}$$

Answer: _____

- 6. Find the mistake.** Marcus simplifies $\frac{20}{3 + i}$. He multiplies the numerator and the denominator by $3 - i$, and then writes the new denominator as 8.

- (a) Explain why 8 is wrong, and give the correct denominator.
- (b) Finish the division correctly and write the result in the form $a + bi$.

Answer: _____

7. Solve. Give both solutions in the form $a + bi$, and state how the two answers are related:

$$x^2 - 6x + 13 = 0$$

Answer: _____

8. Find all four zeros of $P(x) = x^4 - 16$, real and non-real. Check that the non-real zeros you write down form a conjugate pair.

Answer: _____